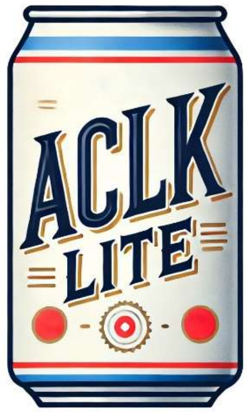
ACLK-Lite



The crisp, refreshing taste of TCLK, packed with data!
All from the modified Manchester you know & love.

ACLK-Lite is a down-converted subset of ACLK events that support forwards-and-backwards compatibility with TCLK. This protocol utilizes the same Manchester encoding as TCLK for all 96b of the ACLK Frame. ACLK-Lite is decoded by a Fermilab provided FPGA IP core that receives either TCLK or ACLK events depending on the external interface provided. Decoding ACLK-Lite in this manner allows for systems under active development to receive either interface (TCLK/ACLK-Lite) without modifications to the FPGA firmware.

ACLK-Lite Encoder Module

The ACLK-Lite Encoder Module (AEM) is the device from which ACLK-Lite will be transmitted. This may take the form of an external TTL interface from Controls hardware (the same as current TCLK transmitters) or a LVDS Timing & Links module that handles distribution at a crate level. The AEM will have two user-configurable run modes:

Legacy TCLK

In this mode all legacy events received over the ACLK link are automatically re-broadcast to Manchester encoding as 8 bit TCLK Events. Legacy events are designated with 0x00ZZ in the upper 8 bits of the event field, and as such no data will be truncated. Data will not be re-broadcast. This operating mode is exactly equivalent to the current TCLK protocol, and users can be assured that legacy systems will continue to operate in the same manner.

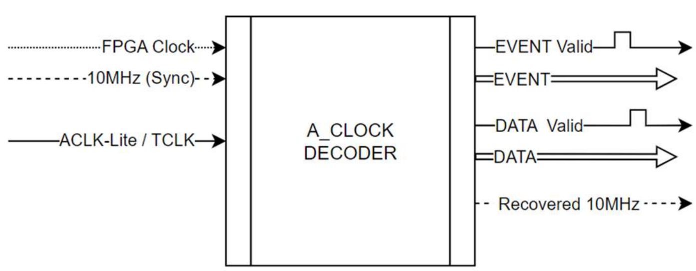
ACLK-Lite

In this mode a subset of events received over the ACLK link are automatically re-broadcast to Manchester encoding as 96 bit ACLK Events with data packet. Up to 16 events may be designated by the user for pass-through to the ACLK-Lite protocol. Encoding data bits to the legacy baud rate incurs additional transmission time. Users should take care to select events based on priority. Collision handling for this mode is covered in the subsequent section.

ACLK-Lite Decoder Module

The ACLK-Lite Decoder Module (ADM) has the following interfaces:

- **FPGA Clock** – A 100~125MHz clock for over-sampling of the Manchester data stream.
- **10MHz Sync**– A 10MHz clock for high precision synchronization of events (optional).
- **ACLK-Lite / TCLK** – The 10MHz Manchester encoded data stream carrying events.
- **EVENT Valid** – A single bit, asserted after the event has been decoded
- **EVENT** – A 16b bus containing the decoded event, valid on EVENT Valid
- **DATA Valid** – A single bit, asserted after the data has been decoded
- **DATA** – A 64b bus containing the decoded event data, valid on DATA Valid.
- **10MHz Recovered** – A 10MHz clock, loosely tied to the TCLK/ACLK stream

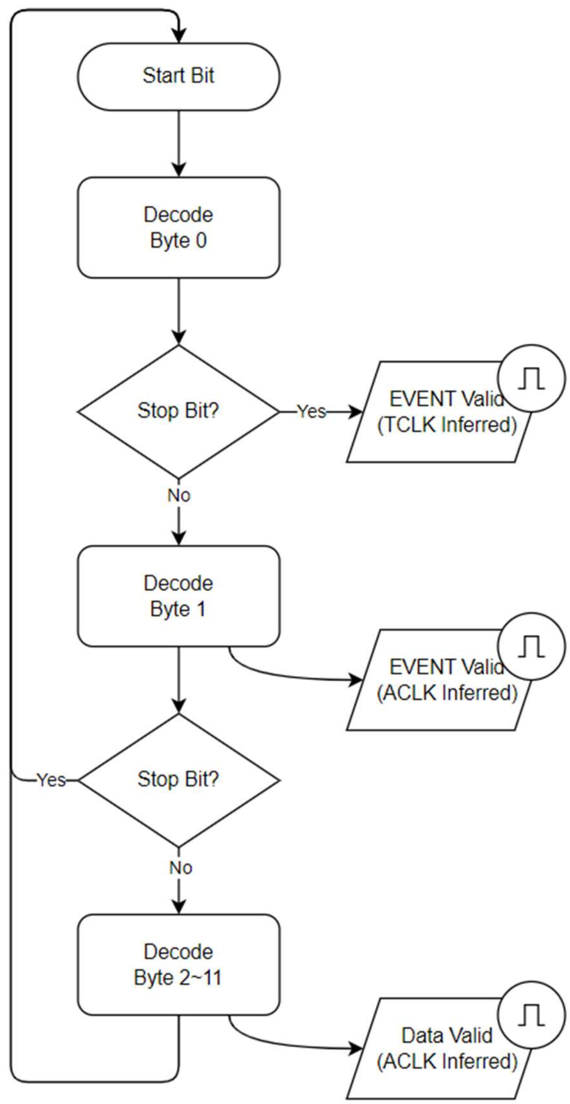


A Clock Decoder Module

Module Behavior

The module decodes TCLK or ACLK-Lite in the following manner (see timing diagrams for more information regarding Manchester encoding).

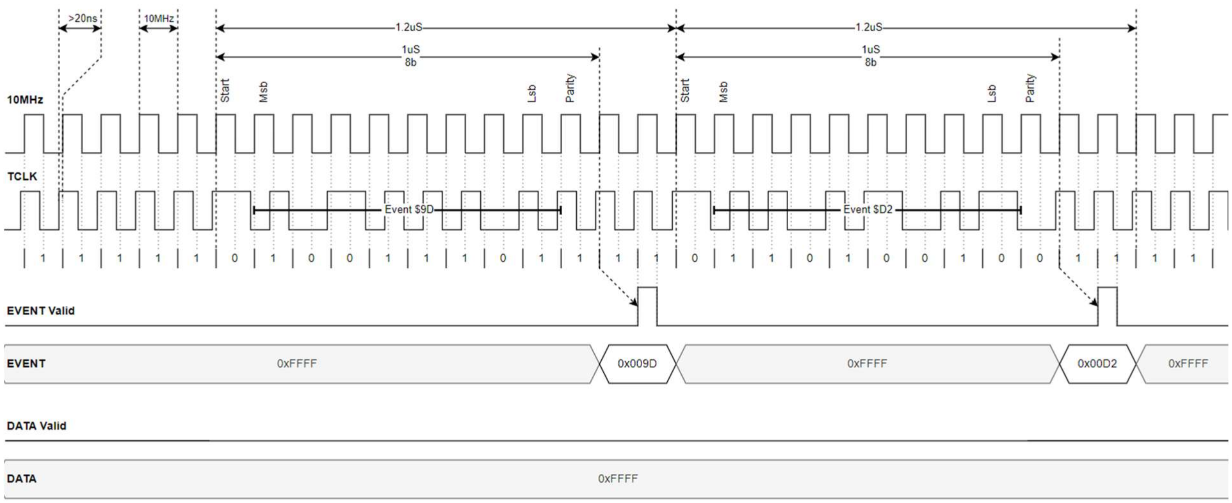
1. On start bit, begin decoding the CLK Event.
2. If the event terminates after 8b, it is a TCLK Event. Output to the EVENT port.
3. If the event terminates after 16b, it is an ACLK Event. Output to the EVENT port.
4. As the event continues, it decodes the ACLK Data. Output to the DATA port.
5. Return to watching for start bits.



Module Behavior

Behavior for Legacy TCLK

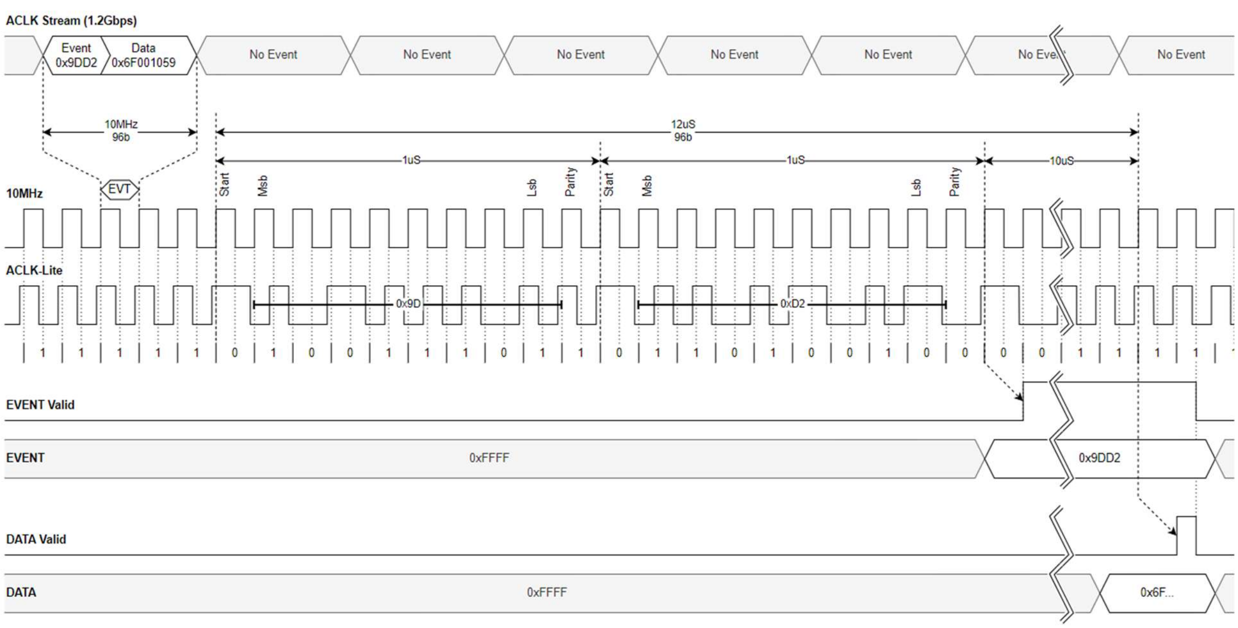
When TCLK is utilized for the ACLK-Lite Decoder Module (ADM), the EVENT port will strobe after 8 bits have been decoded (1.2uS). The upper 8 bits of the EVENT port will be empty. The DATA bus will never be asserted. This functionality is equivalent to current TCLK decoders.



TCLK Decoding

Behavior for ACLK-Lite

When ACLK is utilized for the ADM, the EVENT port will strobe after 16 bits have been decoded (2.2uS). The upper 8 bits of the EVENT port will contain the full ACLK event. The ADM will continue for 10uS until the full data packet has been decoded, at which point it will assert the DATA Valid bit. The EVENT Valid bit will be held high until DATA Valid is de-asserted.



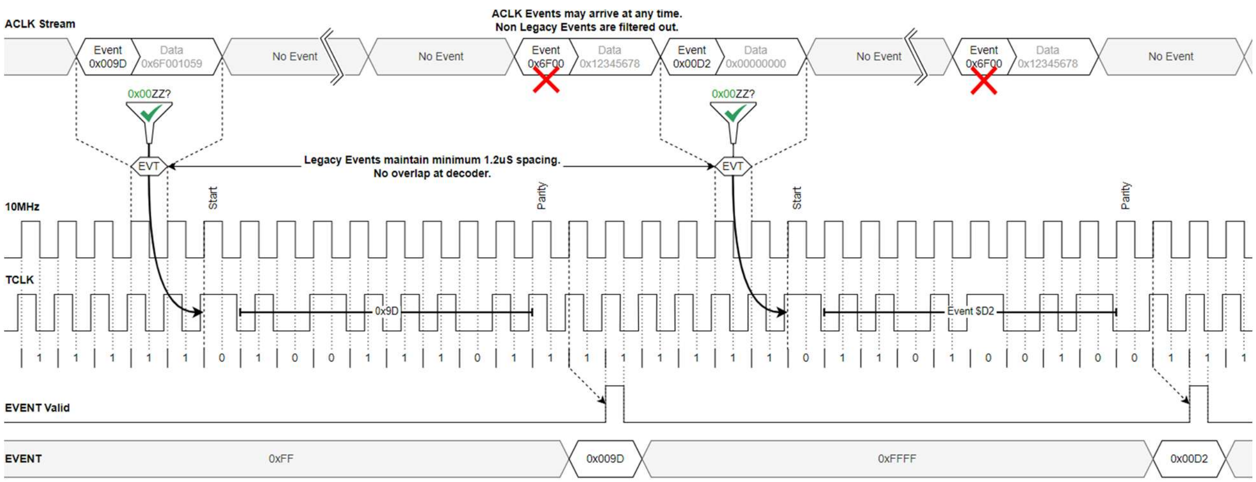
ACLK-Lite Decoding

Collision Handling

Systems wishing to recover ACLK data at higher rates than are supported by the ACLK-Lite protocol should utilize gigabit ACLK. This will likely be the case for precision regulation and trigger systems utilized by Fermilab. For cycle-by-cycle triggers, or periodic acquisition, the ACLK-Lite Encoder Module (AEM) will handle collisions in the following manner.

Legacy TCLK collisions

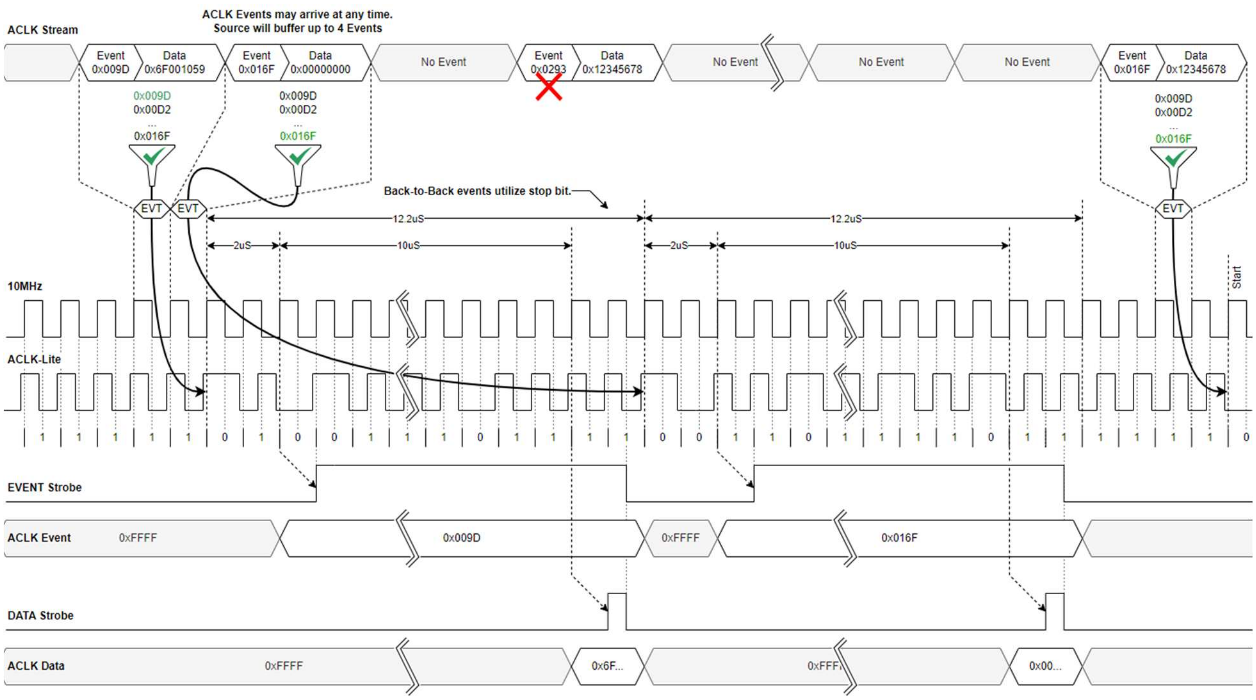
As these events are broadcast to ACLK with the same 1.2uS or greater interval as current TCLK events, collisions will not occur for either the Encoder or Decoder modules. Higher order ACLK events (those utilizing the upper 8 bits) are to be filtered out, in addition to any data contained by the packet. This ensures seamless transition to the new transmitter module.



TCLK Collision Handling

ACLK-Lite collisions

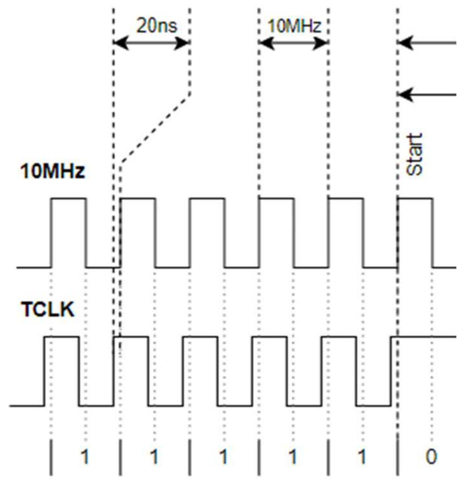
ACLK-Lite incorporates both events at higher rates, and their associated data packets. As such collisions are possible for events broadcast withing 12.2uS of the preceding packet. The AEM will buffer up to 4 events, to be played out concurrently in this scenario. User-defined event filters should take into account the frequency and spacing of events to be decoded. At present, the highest anticipated repetition rate for ACLK events is 5760Hz. This duty factor provides sufficient spacing between ACLK-Lite events that this form of collision will occur infrequently of the downstream system, but is not impossible.



ACLK-Lite Collision Handling

Module Synchronization

The module has an optional 10MHz input to allow precision 10MHz sources or Beam Synchronous clocks to gate events. This ensures <<10ns accuracy for all events on the 10MHz clock domain. Decoding events by oversampling the CLK stream will have at best 10ns accuracy, with a nominal recovered clock jitter of 20ns depending on sampling conditions. CLK data for modules tied to a precision 10MHz source will have a guaranteed 20ns phase advance relative to the rising edge of the reference source.



Other Considerations

The rate and severity of collisions is under active analysis and will be documented along with updated Event information as the ACLK timing system evolves.

Parallel encoding of ACLK packets may be accomplished by crate systems with multiple channels on the backplane. This can be used to send Event and Data packets concurrently.

Alternate encoding of ACLK-Lite packets (8b/10b) at higher rates may be applied for integrated systems, or users wishing to receive ACLK information through standard serial channels.

A third operating mode for the AEM that transmits only event data, thus reducing the 12uS transmit time to 2uS may be able to handle event transmission without filtering.

All of these options are under consideration, but presently out of scope as of March 2025.